

Programming Assignment 2

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Course: Introduction to Blockchain

Instructor: Sir Ahmed Akhtar

Date: 3rd May 2026

Everything has been maintained on my git repository on https://github.com/z-aqib/CSE355_Introduction-to-Blockchain. I have attached the file directory of my folder PA2 as well. I am also attaching the console/terminal proof of all test cases passed successfully.

Proof of test cases passed:

> npx hardhat test

Basic Banking

- ✓ 1_Test
- ✓ 2_Test
- ✓ 3_Test
- ✓ 4_Test
- ✓ 5_Test
- ✓ 6_Test
- ✓ 7_Test
- ✓ 8_Test (61ms)
- ✓ 9_Test (63ms)

Exceptional Cases

- ✓ 1_Test
- ✓ 2_Test
- ✓ 3_Test
- ✓ 4_Test
- ✓ 5_Test
- ✓ 6_Test
- ✓ 7_Test

- ✓ 8_Test
- ✓ 9_Test
- ✓ 10_Test
- ✓ 11._Test
- ✓ 12._Test
- ✓ 13._Test
- ✓ 14._Test
- ✓ 15._Test
- ✓ 16._Test

Interest Banking

- ✓ 1_Test
- ✓ 2_Test (44ms)
- ✓ 3_Test (59ms)
- ✓ 4_Test
- ✓ 5_Test (46ms)
- ✓ 6_Test
- ✓ 7_Test
- ✓ 8_Test

Lock

Deployment

- ✓ Should set the right unlockTime
- ✓ Should set the right owner
- ✓ Should receive and store the funds to lock
- ✓ Should fail if the unlockTime is not in the future

Withdrawals

Validations

- ✓ Should revert with the right error if called too soon
- ✓ Should revert with the right error if called from another account
- ✓ Shouldn't fail if the unlockTime has arrived and the owner calls it

Events

- ✓ Should emit an event on withdrawals

Transfers

- ✓ Should transfer the funds to the owner

Owner Rights

✓ 1_Test

✓ 2_Test

✓ 3_Test

✓ 4_Test

✓ 5_Test

✓ 6_Test

✓ 7_Test

✓ 8_Test

✓ 9_Test

✓ 10_Test

52 passing (3s)

File directory:

```
> Get-ChildItem -Recurse -Force | Where-Object {  
>>   $_.FullName -notmatch '\\(node_modules|\\.git)(\\|$)'  
>> }
```

Directory: D:\Users\DELL\new-onedrive\OneDrive - Institute of Business
Administration\iba\sem8\blockchain\CSE355_Introduction-to-
Blockchain\programming_assignments\PA2

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	03-May-26 09:37 PM		artifacts
d-----	03-May-26 09:37 PM		cache
dar--l	03-May-26 09:29 PM		contracts
d-----	03-May-26 09:29 PM		ignition
d----l	03-May-26 09:21 PM		Solidity Test Files
dar--l	03-May-26 09:29 PM		test
-a----	03-May-26 09:29 PM	270	.gitignore
-a---l	03-May-26 08:52 PM	5576	banking_xxxxx_contract.sol
-a---l	03-May-26 09:20 PM	3199016	CSE355_Spring26_PA2.pdf
-a---l	03-May-26 08:52 PM	1744	grader.py
-a----	03-May-26 09:27 PM	146	hardhat.config.js
-a----	03-May-26 09:45 PM	15449	PA1_zuha_aqib_26106.docx
-a----	03-May-26 09:46 PM	94554	PA1_zuha_aqib_26106.pdf
-a----	03-May-26 09:35 PM	275047	package-lock.json
-a----	03-May-26 09:35 PM	385	package.json
-a----	03-May-26 09:27 PM	390	README.md

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Administration\iba\sem8\blockchain\CSE355_Introduction-to-
Blockchain\programming_assignments\PA2\artifacts

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	03-May-26 09:37 PM		build-info

d----- 03-May-26 09:37 PM contracts

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Blockchain\programming_assignments\PA2\artifacts\build-info

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	03-May-26 09:37 PM	882602	0ea17c13285d266a3e82d28ed187260f.json

Directory: D:\Users\DELL\new-onedrive\OneDrive - Institute of Business
Administration\iba\sem8\blockchain\CSE355_Introduction-to-
Blockchain\programming_assignments\PA2\artifacts\contracts

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	03-May-26 09:37 PM		banking_26106_contract.sol
d-----	03-May-26 09:37 PM		Lock.sol

Directory: D:\Users\DELL\new-onedrive\OneDrive - Institute of Business
Administration\iba\sem8\blockchain\CSE355_Introduction-to-
Blockchain\programming_assignments\PA2\artifacts\contracts\banking_26106_contract.sol

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	03-May-26 09:37 PM	108	BankingSystem.dbg.json
-a----	03-May-26 09:37 PM	64683	BankingSystem.json

Directory: D:\Users\DELL\new-onedrive\OneDrive - Institute of Business
Administration\iba\sem8\blockchain\CSE355_Introduction-to-
Blockchain\programming_assignments\PA2\artifacts\contracts\Lock.sol

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	03-May-26 09:37 PM	108	Lock.dbg.json
-a----	03-May-26 09:37 PM	6552	Lock.json

Directory: D:\Users\DELL\new-onedrive\OneDrive - Institute of Business Administration\iba\sem8\blockchain\CSE355_Introduction-to-Blockchain\programming_assignments\PA2\cache

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	03-May-26 09:39 PM	2139	solidity-files-cache.json

Directory: D:\Users\DELL\new-onedrive\OneDrive - Institute of Business Administration\iba\sem8\blockchain\CSE355_Introduction-to-Blockchain\programming_assignments\PA2\contracts

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a---l	03-May-26 09:29 PM	9087	banking_26106_contract.sol
-a----	03-May-26 09:27 PM	1008	Lock.sol

Directory: D:\Users\DELL\new-onedrive\OneDrive - Institute of Business Administration\iba\sem8\blockchain\CSE355_Introduction-to-Blockchain\programming_assignments\PA2\ignition

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	03-May-26 09:29 PM		modules

Directory: D:\Users\DELL\new-onedrive\OneDrive - Institute of Business Administration\iba\sem8\blockchain\CSE355_Introduction-to-Blockchain\programming_assignments\PA2\ignition\modules

Mode	LastWriteTime	Length	Name
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-a----	03-May-26 09:27 PM	564	Lock.js

Directory: D:\Users\DELL\new-onedrive\OneDrive - Institute of Business
Administration\iba\sem8\blockchain\CSE355_Introduction-to-
Blockchain\programming_assignments\PA2\Solidity Test Files

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a---1	03-May-26 08:52 PM	9655	banking_test.sol
-a---1	03-May-26 08:52 PM	13436	exceptions_test.sol
-a---1	03-May-26 08:52 PM	15592	interest_test.sol
-a---1	03-May-26 08:52 PM	7965	owner_test.sol

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Blockchain\programming_assignments\PA2\test

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a---1	03-May-26 08:52 PM	7030	banking_test.js
-a---1	03-May-26 08:52 PM	5641	exceptions_test.js
-a---1	03-May-26 08:52 PM	7018	interest_test.js
-a----	03-May-26 09:27 PM	4317	Lock.js
-a---1	03-May-26 08:52 PM	3084	owner_test.js